Reflection (Jagadeesh Bhaskara):

I designed the system as a step-by-step research pipeline so that each agent had a distinct job. The first agent was responsible for expanding the original question into a clearer research plan, with sub-questions, keywords, and possible search directions. I chose this setup because a broad research question is often too vague to search effectively on its own. The second agent was meant to gather evidence from the paper corpus, while the third agent focused on identifying major themes, overlaps, and disagreements across the sources. The final agent then turned all of that into a structured literature review.

This design worked well because it mirrored how I would approach a literature review myself: first understand the question, then gather evidence, then compare and interpret the sources, and finally write the report. Giving each agent a specific role also made the workflow easier to follow and helped keep the output more organized.

I also tried to make the reasoning process visible by encouraging the agents to use a ReAct-style approach. In practice, this meant that the agents would first think through the problem, then take an action such as searching for relevant passages or comparing themes and then reflect on what they found. For example, the query expander might identify that the question is too broad and decide to break it into smaller parts. The source hunter might realize that the first search results are too general and refine the search to focus on more specific concepts. The synthesizer might then notice that the literature agrees on one point but differs on another, which helps shape the final analysis. These steps made the reasoning process feel more grounded.

Overall, the system worked reasonably well. The separation of tasks made the workflow clear, and the final reports were more structured than they would have been with a single agent. The retrieval step was especially important, because the quality of the sources strongly affected the quality of the synthesis and writing. At the same time, there were some limitations. The agents sometimes produced outputs that were slightly too general, and the retrieval step did not always find the most precise or relevant passages. If I were to improve the system, I would make the prompts more specific and add some checks to help filter out weak or off-topic results before the final report is written.

To use this system in production, I would need stronger retrieval methods, better evaluation tools, and more safeguards around accuracy. In particular, it would help to have a way to verify that the reported claims are supported by the papers, rather than relying only on the model's own generation. A production version would also benefit from better monitoring, more robust prompt design, and perhaps a human review step for important outputs. As a prototype, the system is useful and demonstrates the value of multi-agent reasoning, but it would still need further refinement before it could be trusted for high-stakes research work.